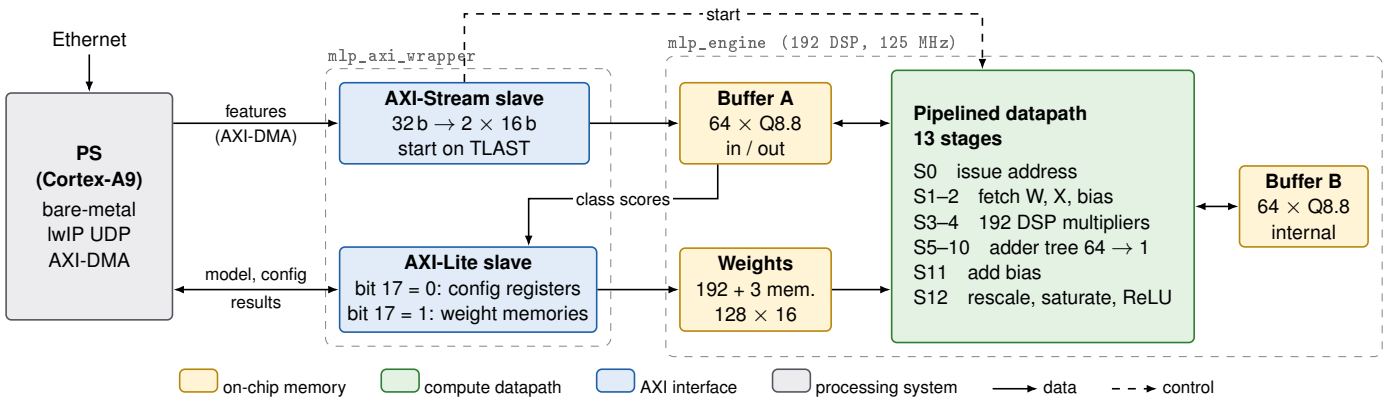
Ethernet
PS
(Cortex-A9)
bare-metal
lwIP UDP
AXI-DMA
features
(AXI-DMA)
model, config
results
mlp_axi_wrapper
AXI-Stream slave
32 b → 2 × 16 b
start on TLAST
AXI-Lite slave
bit 17 = 0: config registers
bit 17 = 1: weight memories
start
mlp_engine (192 DSP, 125 MHz)
class scores
Buffer A
64 × Q8.8
in / out
Weights
192 + 3 mem.
128 × 16
Pipelined datapath
13 stages
S0 issue address
S1–2 fetch W, X, bias
S3–4 192 DSP multipliers
S5–10 adder tree 64 → 1
S11 add bias
S12 rescale, saturate, ReLU
Buffer B
64 × Q8.8
internal
on-chip memory
compute datapath
AXI interface
processing system
data
control